



Awesome-ai4sci-papers

一个精选的 **AI for Science** 领域文献清单，持续更新中。

Foundation Models

基于大规模生物组学数据预训练的基础模型。

Year / Venue	Paper	Tags
2024 <i>Nature Methods</i>	scGPT: Toward Building a Foundation Model for Single-Cell Multi-Omics Using Generative AI [paper] [GitHub]	Single-cell omics , Multi-omics

AI Agents

具备自主规划、工具调用与反思迭代能力的 LLM 智能体，完成生物数据分析任务。

Year / Venue	Paper	Tags
2026 <i>Nature</i>	Accelerating Scientific Discovery with Co-Scientist [paper]	Cross-Modal , Scientific Discovery
2026 <i>Nature</i>	An AI System to Help Scientists Write Expert-Level Empirical Software (ERA) [paper]	Tools & Infrastructure , Pipeline Optimization
2026 <i>Nature</i>	A Multi-Agent System for Automating Scientific Discovery (Robin) [paper]	Cross-Modal , Scientific Discovery
2026 <i>Nature Methods</i>	CellVoyager: AI CompBio Agent Generates New Insights by	Single-cell omics , Scientific Discovery

Year / Venue	Paper	Tags
	Autonomously Analyzing Biological Data [paper]	
2026 <i>ICLR</i>	CELLAGENT: LLM-Driven Multi-Agent Framework for Natural Language-Based Single-Cell Analysis [paper] [GitHub]	Single-cell omics , Pipeline Optimization
2025 <i>Nature Communications</i>	CASSIA: A Multi-Agent Large Language Model for Automated and Interpretable Cell Annotation [paper]	Single-cell omics , Pipeline Optimization

Benchmarks

该领域的系统性综述与基准评测研究。

Year / Venue	Paper	Tags
2022 <i>Nature Methods</i>	Benchmarking Atlas-Level Data Integration in Single-Cell Genomics [paper]	Single-cell omics

Tools & Infrastructure

广泛使用的生物信息学工具与计算基础设施。

Year / Venue	Paper	Tags
2018 <i>Genome</i>	SCANPY: Large-Scale Single-Cell Gene Expression Data Analysis [paper] [GitHub]	Single-cell omics , Tools & Infrastructure

Year / Venue	Paper	Tags
Biology		

Tags 说明

Tag	Description
Single-cell omics	应用于单细胞组学数据分析，包括 scRNA-seq、scATAC-seq、scMulti-omics 等
Spatial omics	应用于空间组学数据分析，包括空间转录组、空间蛋白质组等
Multi-omics	模型设计整合了多种组学层次（如转录组 + 蛋白组 + 表观组）
Bioimage	应用于生物学图像分析，包括细胞图像、显微图像等
Cross-Modal	Agent 能力跨越多个模态或领域，非单一组学任务
Scientific Discovery	核心能力为自主提出科学假设、设计实验方案、推动生物学新发现
Pipeline Optimization	核心能力为自主构建、优化、执行端到端生物信息学分析流程
Tools & Infrastructure	经典生物信息学工具或计算基础设施，具有广泛的应用价值